

Detection Is Not Edge: A Leakage-Free, Cost-Aware Audit of LLM Agents, Regime Models, and Candlestick Motifs for Stock-Movement Onset on Chinese A-Shares

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ABSTRACT

We report a deployment-realistic audit of machine-learning and large-language-model (LLM) methods for *movement-onset detection* — predicting the *start* of a price move — on the Chinese A-share market, anchored to a deployed production system (V12.31; documented α +2.236pp/month, Sharpe 2.20). We package the production logic as **V12-Agentic**, a four-agent framework (Macro Regime Monitor, Alpha Factor Explorer, Pattern Core, Backtest & Verifier), and a knowledge-encoding methodology that converts tacit operator expertise into a structured agent prompt. Against this framework we run a rigorous, mostly negative evaluation on a 5.27M-anchor panel under a strict leakage-free walk-forward, realistic A-share transaction costs, date-clustered bootstrap confidence intervals (CIs), and the Deflated Sharpe Ratio (DSR). Six findings follow. (C1–C2) the knowledge-encoding procedure and the agentic framework; (C3) apparent hybrid-over-LGBM gains are *within date-clustered noise*; (C4) a quarter-conditional oracle bounds any router at +0.39pp Top-10%; (C5) on a leakage-free substrate the *identified* LLM contribution to onset is ≈ 0 (raw +0.033, expert +0.006; both span zero), LLM-as-weak-supervisor is harmful (−0.149), and a cross-market de-biasing estimator collapses public benchmark “skill” (67–80%) to chance; (C6) candlestick onset information is real, regime-conditional, cross-period-stable, and even directionally monotone, yet its gross edge ($\approx 0.1\%/5d$) is *below the transaction-cost floor* — information, not tradable alpha. Reimplementing three 2025–26 SOTA baselines (regime LightGBM, two-level abstention, NMI information theory) under our protocol, *none survives*. We connect these nulls to two architectural limits of LLMs — proactive interference and mode collapse. Across 11 research lines and 60+ method/arm comparisons, no configuration yields deployable alpha from public features; the contribution is the honest evaluation machinery and a precise localization of *why* detection $\neq$ edge.

KEYWORDS

stock movement prediction, movement onset, LLM agents, data leakage, transaction costs, mutual information, regime models, honest evaluation, A-shares

1 INTRODUCTION

Stock-movement prediction has been studied for decades, yet a persistent gap separates academic methods from deployed systems. Academic work typically evaluates on stratified subsets that over-represent the event of interest (e.g. 25% positive class), while deployment must operate under the true marginal ($<10\%$). Recent

LLM agents [9, 14, 28] intensify the gap: whether they add genuine alpha in a production pipeline, and how to combine them with mature tabular models, is poorly studied at scale. The cost of producing a beautiful-but-worthless backtest with AI assistance is now near zero, which accelerates over-fitting and false discovery rather than solving research.

We address this with **V12-Agentic**, a four-agent framework grounded in a deployed A-share system (V12.31), and subject it to a deliberately adversarial, leakage-free, cost-aware evaluation. The A-share setting is an ideal testbed: (i) most retail-accessible stocks *cannot be shorted*, so a bearish signal is an avoidance criterion, not a tradeable direction; (ii) onset events are sparse ($\approx 8\%$); (iii) recurring “disaster months” induce regime non-stationarity.

Contributions. This paper makes six contributions. The methodological novelty is C1, C2, and the leakage-free *identification* framework C5; C3–C4 and C6 are honest, deployment-realistic empirical findings rather than new techniques.

- **C1 — Knowledge encoding.** A procedure converting tacit production knowledge (onset rules, disaster signals, asymmetric short-sale constraints) into a $\sim 5,500$ -character structured agent prompt (§4, App. A).
- **C2 — V12-Agentic architecture.** A four-agent decomposition mapping to a real production codebase (§4).
- **C3 — Hybrid effects are within noise.** A stratified “+54% over LGBM” reverses to -15% on a random sample; three controlled experiments show this is *not* a base-rate mechanism but date-clustered noise: the headline gap is +0.72pp, 95% CI $[-0.56, +2.05]$ pp (§5.1).
- **C4 — Quarter-conditional headroom.** An oracle upper bound of +0.39pp Top-10% for any online regime router; our routers capture ≈ 0 of it (§5.2).
- **C5 — Leakage-free identification.** Treating leakage-freeness as an identification condition, the LLM’s incremental contribution to A-share onset is ≈ 0 ; weak supervision is harmful; a de-biasing estimator collapses contaminated-benchmark skill to chance (§5.4).
- **C6 — Detection is not edge.** Candlestick onset information is real, regime-conditional, cross-period-stable, and directionally monotone, yet sub-cost (§5.5); reimplemented SOTA regime/abstention/information-theory baselines are *also* sub-cost under our protocol (§6).

Figure 1 previews all 11 lines and 60+ methods compared.

stockagent-research — A-share movement-onset detection

60+ methods / arms compared across 11 lines (green = robust positive · amber = in-sample / within-noise · red = null / sub-cost / not-deployable · gray = shelved)

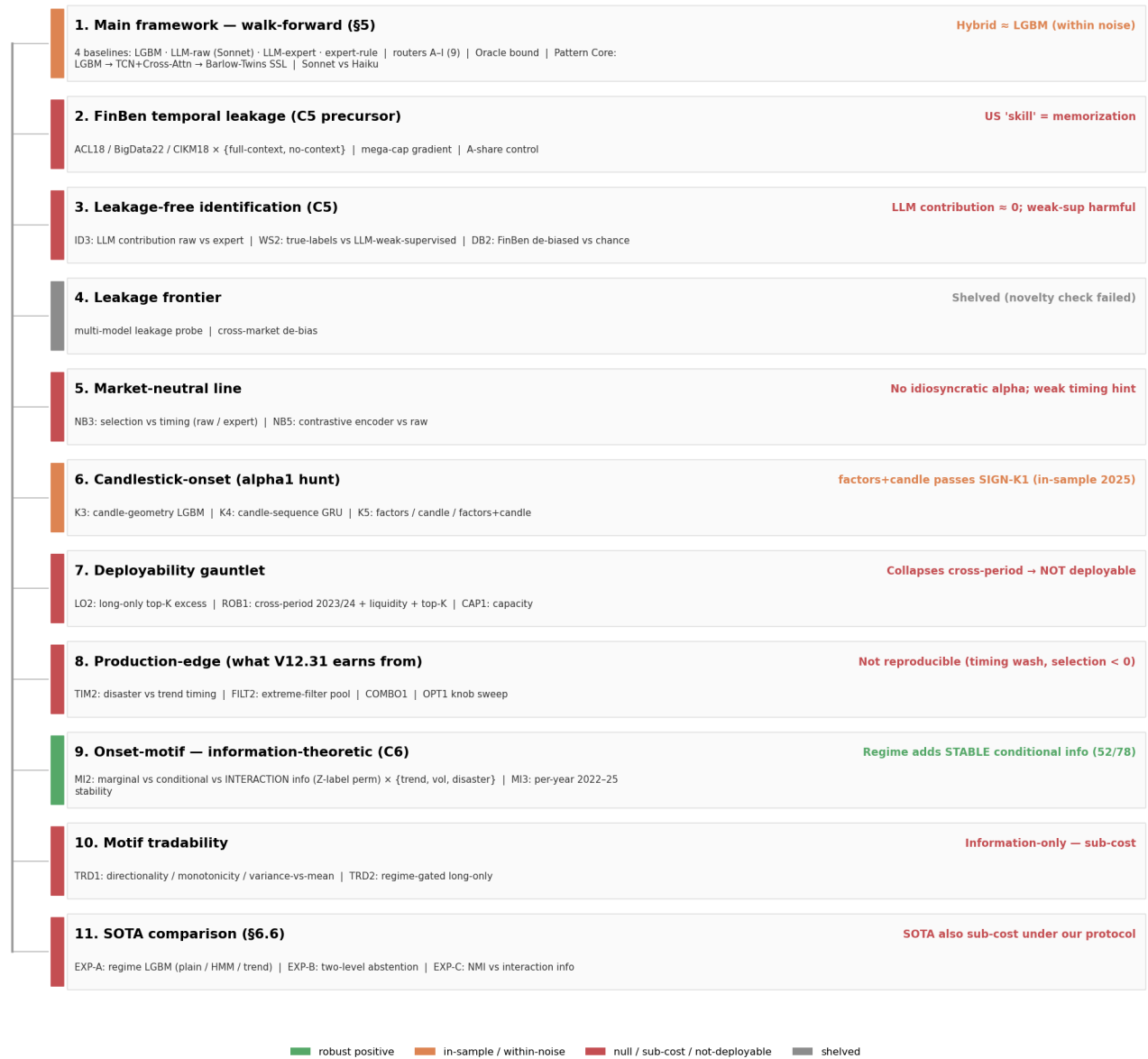


Figure 1: Overview of every method/arm compared across the project’s 11 research lines, colour-coded by verdict (green: robust positive; amber: in-sample / within-noise; red: null / sub-cost / not deployable; gray: shelved). Across 60+ comparisons, only the onset-motif *information* (C6) is a robust positive, and it is sub-cost.

2 RELATED WORK

Stock movement prediction. Triple-barrier labeling [15], graph and attention models (HIST [27], MASTER [13], StockMixer [6], FactorVAE [5]), and gradient boosting [7, 11] are standard. Recent work questions optimal supervision horizons (Label Horizon Paradox) and the practicality of chart-based deep learning, the latter

concluding that DNN chart predictors create false positives once temporal context is respected — an independent echo of our C5/C6.

LLM agents in finance. FinAgent [28], AlphaForge [9], FinRobot [26], TradingGPT [14] and domain-tuned LLMs evaluate mostly on curated benchmarks. A growing line documents temporal leakage and “profit mirages” in LLM-finance [8, 12, 16, 22, 24].

Architectural limits of LLMs as alpha generators. Two recent results give a mechanistic basis for why our identified LLM contribution is null rather than merely small. *Proactive interference* [25] shows LLM retrieval of the latest value in a stream of related updates decays log-linearly toward zero as prior updates accumulate — a working-memory bottleneck independent of context-window size and not fixed by prompting, across 30+ models. This undercuts feeding long multi-period histories to an LLM and expecting it to track the *current* state. *Mode collapse* [10] (NeurIPS 2025 Best Paper) documents an “Artificial Hivemind” across 70+ models: open-ended generations converge both within a model and across families, so neither temperature nor ensembling restores diversity. Because alpha lives in the long tail, a mode-seeking generator structurally reproduces already-arbitraged “traditional wisdom.” These limits frame C5 and C6 as expected consequences of using an LLM as ideator/predictor rather than as a rigorous execution-and-falsification engine under human-supplied hypotheses.

Information theory, transaction costs, multiple testing. We use interaction information [17] and conditional mutual information [4] with permutation inference [21]; the anomaly-cost-erosion literature [20] and regime-conditional returns [3] contextualize C6; and we report the Deflated Sharpe Ratio [2] to correct for the many strategy variants tried.

3 MOVEMENT ONSET: PROBLEM AND ASYMMETRY

For (i, t) with adjusted close $P_{i,t}$, let $r_{i,t}^{(h)} = P_{i,t+h}/P_{i,t} - 1$.

Definition 1 (Bullish onset). A bullish onset occurs at (i, t) iff (1) forward magnitude $r_{i,t}^{(H)} \geq \Delta^+$; (2) bounded drawdown $\min_{0 \leq h \leq H} P_{i,t+h}/P_{i,t} \geq 1 - \Delta^-$; (3) backward context $\sum_{s=t-W}^{t-1} \log(P_{i,s+1}/P_{i,s}) \leq \tau$. In V12.31, $(H, \Delta^+, \Delta^-, W, \tau) = (5, 0.10, 0.05, 5, 0.08)$. Condition (3) prevents labeling already-trending stocks as onsets.

Asymmetric short-sale constraint. Most A-shares cannot be shorted, so a bearish onset can only *remove* a stock from a long-only pool (Bearish-as-Filter), and negative peer information carries disproportionate predictive weight [3]. This motivates a two-track design: a long-bullish track plus a bearish-onset filter.

Sparse events and sampling. The bullish onset rate is $\approx 8.0\%$. Three regimes appear: stratified-PoC (25% positive), walk-forward-random (7.9–9.4%, the deployment distribution), and quarter-stratified. We show that stratified-PoC overstates hybrid advantage (§5.1).

Evaluation targets. RankIC (per-date cross-sectional Spearman vs $r^{(5)}$), Top-K return and winrate ($K \in \{10\%, 20\%\}$), with 1000-resample date-clustered bootstrap CIs. For tradability we additionally report long-only top-K market-excess net of realistic A-share round-trip cost, annualized Sharpe, and DSR.

4 V12-AGENTIC ARCHITECTURE

Four agents operate in a producer–consumer cascade.

Table 1: Pattern Core phases (mean over 3 walk-forward splits). Phase 3 matches Phase 1 with 116× less labeled data and no deployment-time LLM cost.

Phase	Labeled data	RankIC	Top-10%
Phase 1: LGBM	3.5M anchors	+0.088	+1.63%
Phase 2: TCN	100K (35× less)	+0.060	+1.61%
Phase 3: BT-SSL+TCN	30K lab. + 100K unlabeled	+0.077	+1.63%

Macro Regime Monitor. A daily binary disaster flag $z_t^M = \mathbb{1}[(A_t + B_t + C_t) \geq 2]$ from index-AND (A), volume-OR (B), and sector inner-vote (C) signals. On D1 it flags 12/1062 days (1.13%).

Alpha Factor Explorer. Maintains 153 technical/fundamental factors plus 8 concept-heat factors; ten forward-looking columns ($r5-r40$, $dd5-dd40$; $\text{spearman}(r5, r^{(5)}) = 0.898$) are blacklisted, leaving 165 effective features. Its core asset is the **expert-knowledge prompt** (C1): a four-round interview elicitation encoding the bullish-onset rules (C1 bottoms-rising, C2 above-5d-low, C3 MA pattern, C4 volume), the bearish asymmetry, and the disaster composite. The executable rule fires at 7.76%, matching the 8% production label.

Pattern Core. Emits per-stock onset probabilities. *Phase 1* is Light-GBM (63 leaves, 500 rounds, leakage-protected). *Phase 2* is a TCN-causal encoder (dilations $\{1, 2, 4, 8\}$, $d=64$) with bidirectional spatio-temporal cross-attention (4 heads; 119,171 params). *Phase 3* pre-trains that encoder with Barlow Twins [29] SSL on unlabeled anchors then finetunes with differential learning rates. Table 1 compares phases.

Backtest & Verifier. Combines Pattern Core and LLM signals via nine routing strategies (A–I: confidence, stratum, soft-ensemble, avoid-expert, LGBM-floor-LLM-boost, disaster-aware, disagreement-boost, regime-ensemble, onset-aware), then runs walk-forward evaluation with bootstrap CIs.

5 EXPERIMENTS

Dataset. **D1** (Tushare Pro, A-shares 2022-01-04 to 2026-05-27): 5,275,812 anchors, 5,434 stocks, 1,062 trading days, 185 columns; onset rate 8.0%. Walk-forward splits use expanding train windows with test windows 2025Q2/Q3/Q4. LLM evaluation uses `claude-sonnet-4-6`; the full 12,000-call walk-forward cost \$27.

5.1 Main results and the within-noise finding (C3)

Table 2 reports the 12 methods. Pure LLM baselines have negative RankIC and far lower Top-10% than LGBM; the best hybrid (G_disagreement_boost) only *ties* LGBM (+1.63%). A stratified PoC shows “hybrid +54% over LGBM” (Table 3) that reverses to −15% on the random sample. Three controlled experiments on the already-scored anchors show the effect is not a base-rate mechanism: (i) sweeping the random pool’s onset-rate/expert-score composition never reproduces hybrid > LGBM (Fig. 2); (ii) swapping in a stronger base ranker leaves the gap unchanged; (iii) under a *date-clustered* bootstrap the headline gap is +0.72pp, 95% CI $[-0.56, +2.05]$ pp

Table 2: Walk-forward results (6,000 anchors; mean $\pm$ std over 3 splits).

Method	RankIC	Top-10%	WR	Top-20%
BL_LGBM	+0.088	+1.63%	60.7%	+1.45%
BL_LLM_raw	-0.073	+0.36%	45.3%	+0.49%
BL_LLM_expert	-0.103	+0.18%	44.6%	-0.05%
BL_expert_rule	-0.089	+0.00%	—	-0.07%
A_confidence	+0.074	+1.51%	57.7%	+1.37%
B_stratum	+0.033	+1.06%	52.2%	+1.07%
C_soft_w0.6	+0.040	+0.85%	51.5%	+0.78%
D_avoid_expert	-0.080	+0.24%	44.8%	+0.27%
E_floor_boost_.15	+0.075	+1.44%	57.5%	+1.41%
E_floor_boost_.30	+0.060	+1.24%	54.8%	+1.17%
F_disaster_aware	+0.060	+1.23%	58.5%	+1.22%
G_disagree_boost	+0.079	+1.63%	61.3%	+1.43%
H_regime_ensemble	+0.047	+1.21%	55.7%	+1.21%
I_onset_aware	+0.069	+1.35%	57.3%	+1.39%

Table 3: Stratified PoC vs walk-forward random (Top-10%). The apparent reversal is the difference of two estimates each indistinguishable from zero for the hybrid-vs-LGBM contrast (C3).

Method	Strat. (25%)	Random (8%)	Δ
BL_LGBM	+1.25%	+1.56%	+0.31
BL_LLM_raw	+0.91%	+0.32%	-0.59
BL_LLM_expert	+1.16%	+0.26%	-0.90
E_floor_boost_.30	+1.93%	+1.32%	-0.61

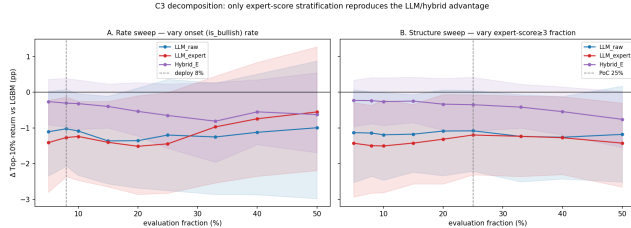


Figure 2: C3 dose-response: sweeping the evaluation pool’s onset-rate / expert-score composition, the hybrid’s Top-10% stays below LGBM at every composition (no zero-crossing). Base rate is not the cause.

— spanning zero. At sample sizes typical of LLM-finance studies, hybrid-over-baseline effects of this magnitude are within date-clustered noise; ratio-reported “LLM helps” claims without cluster-robust CIs are unreliable.

5.2 Oracle headroom (C4) and cross-quarter heterogeneity

Three different methods are optimal on the three splits (Split 1 LGBM +3.34%; Split 2 LLM-raw +1.10% where LGBM fails at RankIC -0.015; Split 3 hybrid +1.48%), yielding a quarter-conditional

Table 4: Directional accuracy on the FinBen stock-movement suite. No-context $\geq$ full-context everywhere: the model predicts better with no data — it is recalling.

Benchmark	n	Base	LLM ctx	LLM no-ctx	mega
ACL18 ('14–16)	3720	49.1	67.2	73.3	74.0
BigData22 ('19–21)	1472	55.2	60.6	71.9	73.5
CIKM18 ('17)	1143	53.4	78.8	80.3	95.4
A-share (control)	1000	50.3	—	48.6	—

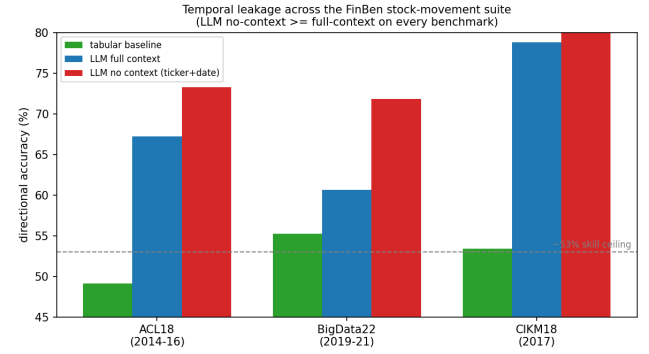


Figure 3: Temporal-leakage signature across the three FinBen benchmarks: no-context accuracy meets or exceeds full-context, with a mega-cap gradient; the A-share control sits at chance.

oracle of +2.02% mean Top-10% vs LGBM +1.63% — a +0.39pp upper bound for *any* online router. Our four regime-aware routers capture ≈ 0 of it; all hybrid-vs-LGBM CIs straddle zero. Phases 2–3 (Table 1) instead bridge the Split-2 LGBM failure (Phase 3 RankIC +0.058, Top-10% +1.72%, +95% over LGBM) at 116 $\times$ less labeled data and zero deployment LLM cost.

5.3 Temporal leakage on public benchmarks (C5 precursor)

We confirm (claiming no novelty) that modern LLMs leak temporally on pre-cutoff data. On the FinBen/PXIU stock-movement suite (Table 4, Fig. 3), gemini-3.5-flash scores 67–80% directional accuracy, but *removing all input* (only ticker+date) raises accuracy on every benchmark, with a strong mega-cap gradient (up to 95.4%). The identical no-context protocol on barely-memorized A-shares falls to chance (48.6%, CI [45.4, 51.7]) — a cross-market control proving the US “skill” is recall, not forecasting.

5.4 Leakage-free identification of the LLM’s contribution (C5)

Leakage-freeness is an *identification condition*: where the no-context probe is at chance (A-shares), the memory channel is null, so any incremental signal is identified as reasoning. (i) The identified LLM contribution over LGBM is +0.033 ([-0.037, +0.110]) raw and +0.006 ([-0.062, +0.078]) expert — both span zero; the first identified (not

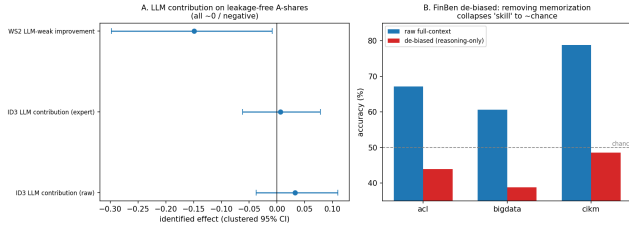


Figure 4: Leakage-free identification (C5): the identified LLM contribution to A-share onset is ≈ 0 ; weak supervision is harmful; de-biased FinBen “skill” collapses to chance.

memory-confounded) estimate, and it is ≈ 0 . (ii) LLM-as-weak-supervisor *hurts*: true-labels RankIC +0.109 vs LLM-refined -0.040 , an identified change of -0.149 ($[-0.298, -0.009]$). (iii) A de-biasing estimator (full – (no-ctx – chance)), valid because the clean market sits at chance, collapses FinBen to reasoning-only ACL18 0.440 / BigData22 0.388 / CIKM18 0.485 — at or below chance. Figure 4 summarizes.

5.5 Detection is not edge: the onset motif (C6)

The candlestick “onset” (relative-position geometry of recent vs prior bars) is the strongest cross-sectional signal in the project, but a deployability gauntlet kills it: the factors+candle long-only edge that looks good on 2025 (net Sharpe 1.10, 3/3 splits positive) is *negative* in 2023 (-0.37) and 2024 (-0.36) — period-specific, not deployable (Fig. 7). Likewise, the documented V12.31 production rules do not reproduce the Sharpe-2.2 edge (timing is a wash, selection is robustly negative across all knobs and OOS).

Is the signal merely conditional? A natural rescue is that the onset is informative only *given* the regime (a “promoter” transcribed only with the right “factor”). We test this information-theoretically. The marginal mutual information $I(\text{pattern}; r^{(5)}) \approx 0$. The correct test for “the regime adds information” is the interaction information $\Pi = I(X; Y | Z) - I(X; Y)$ with a Z -label permutation null (permuting the target within strata only tests the trivial $I(X; Y | Z) > 0$; at $n=2 \times 10^5$ that p floors at $1/n_{\text{perm}}$). On candlestick features $\times \{\text{trend, vol, disaster}\}$ regimes, 52/78 feature $\times$ target $\times$ regime cells have $\Pi > 0$; the trend regime contributes ≈ 70 –76% of the conditional MI, and this is positive and permutation-significant in *every* year 2022–25 (Table 5, Fig. 5).

But MI is sign-blind. Decomposing into conditional mean vs variance shows a *perfectly monotone* directional core (monotonicity coefficient ≈ 1.0 , rank-slope 0.083–0.102), yet only ≈ 37 –43% of the dispersion is in the mean. The decisive test — the simplest regime-gated long-only top-decile portfolio, net of $\approx 0.2\%$ A-share round-trip — yields a *negative net Sharpe in every year* despite gating improving over always-on (Fig. 6). The gross edge ($\approx 0.1\%/5d$) sits below the cost floor. The onset motif is real, regime-conditional, cross-period-stable, directionally monotone information — but not tradable alpha. The binding constraint is transaction cost, not absence of signal, sign blindness, or non-stationarity.

Table 5: Onset-motif interaction information (nats) by year, leading hits. Positive and permutation-significant every year; absolute magnitude is small (correlation-equivalent ≈ 0.07 –0.14).

Feature	Regime	2022	2023	2024	2025
close_pct_prior	trend	+0.0060	+0.0023	+0.0103	+0.0054
dist_low_atr	trend	+0.0059	+0.0021	+0.0091	+0.0056
onset_score	trend	+0.0042	+0.0022	+0.0074	+0.0033

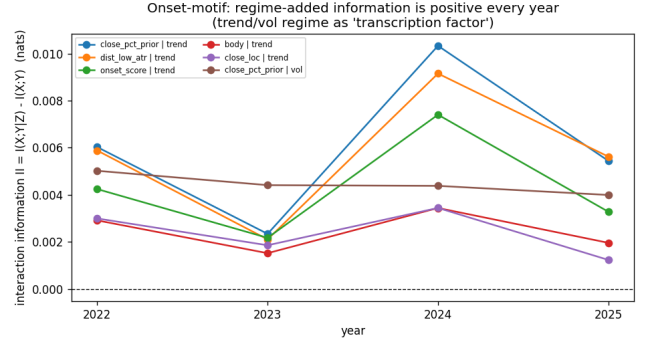


Figure 5: Regime-added information (interaction information) is positive every year 2022–25 for all leading hits.

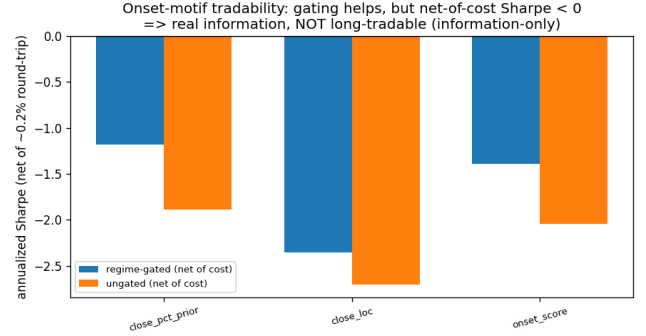


Figure 6: Onset-motif tradability: regime gating helps, but the net-of-cost Sharpe is negative — real information, not long-tradable alpha.

6 ABLATIONS AND SOTA BENCHMARK

Beyond boost-magnitude, model-size (Sonnet vs Haiku), expert-prompt, sample-size, and TCN-scaling ablations, we ask the decisive question: do recent SOTA regime/uncertainty baselines survive our protocol? We reimplement three closest 2025–26 baselines on D1 under our exact protocol — leakage-free walk-forward, A-share cost, non-overlapping 5-day periods, date-clustered CIs, and DSR. We do *not* reproduce their US numbers (different data/task, mostly no code); the fair test is whether their *methods* survive *our* evaluation.

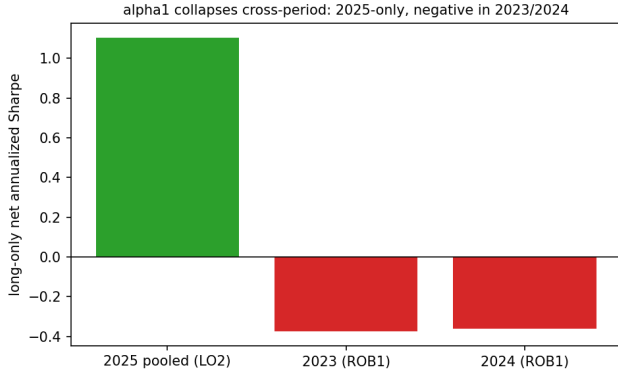


Figure 7: Deployability gauntlet: the 2025 long-only candlestick edge collapses cross-period (2023/2024 negative).

Table 6: SOTA baselines under our protocol (pooled, net of A-share cost). DSR = Deflated Sharpe; CI = date-clustered mean-excess. No arm has a strictly positive mean CI.

Exp	Arm	Sharpe	DSR	mean CI
A	plain LGBM	+0.89	0.65	[−.0016, +.0036]
A	HMM-gated	+1.92	0.99	[0.000, +.0025]
A	trend-gated	+1.45	0.82	[−.0008, +.0038]
B	always-on	−2.04	0.00	[−.0086, −.0031]
B	trend-gate	−1.39	0.01	[−.0046, −.0008]
B	abstention	−1.89	0.00	[−.0075, −.0024]

- **EXP-A — Regime-Aware LightGBM** [1] (rolling-HMM gate). One LGBM ranker, three gates. The HMM arm is the only one to clear the DSR (0.99), but its mean-excess CI lower bound is exactly 0.0: its high Sharpe is a *cash-during-volatility* artifact (gating to cash shrinks the Sharpe denominator), not selection alpha. No arm has a strictly positive net-of-cost mean CI.
- **EXP-B — Two-level abstention** [23] on our onset ranker. All arms net-negative; abstention trades 75% of periods and stays firmly sub-cost. It does not rescue.
- **EXP-C — Information-theory framings** [18, 19]. Marginal NMI ≤ 0.0009 — below the < 0.05 “efficient-market” band — so NMI / conditional-MI framings would call these features signal-free; only our interaction-info probe isolates regime-added information (6/6 hits).

Table 6 and Figure 8 summarize: reimplemented on A-shares under honest evaluation, the closest SOTA methods are sub-cost — like our own signals. Apparent regime Sharpe gains are timing/risk reduction, not net-of-cost alpha.

7 DISCUSSION

Why nothing trades. Across 60+ comparisons (Fig. 1), the only robust positive is the onset-motif *information* (C6), and it is sub-cost. The binding constraint is transaction cost; regime conditioning buys Sharpe via cash-during-volatility risk reduction, not alpha. This converts the thesis from “our signal does not trade” into

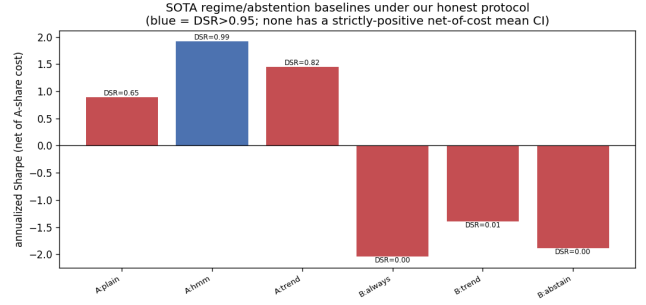


Figure 8: SOTA regime/abstention baselines under our honest protocol (blue: DSR>0.95). None has a strictly positive net-of-cost mean CI.

the stronger “even 2025–26 SOTA methods, on this substrate under honest evaluation, do not trade.”

Human–AI division of labor. The two LLM architectural limits [10, 25] imply a clean boundary: humans supply the edge hypothesis (“who pays you, and why?”); the AI’s value is rigorous execution and adversarial falsification (estimators, cost models, permutation nulls, walk-forward, DSR). Our onset-motif probe arose exactly this way — a human analogy plus an AI-built information-theoretic test — not from asking an LLM to “find alpha.”

Limitations. Single market (A-shares); three quarterly test splits (mitigated by date-clustered CIs); a single LLM family for the agent experiments; public features only — we make no claim about proprietary microstructure data.

8 CONCLUSION

We presented V12-Agentic and a leakage-free, cost-aware audit of LLM and regime methods for A-share movement-onset detection. The identified LLM contribution is ≈ 0 ; candlestick onset information is real, regime-conditional, and cross-period-stable yet sub-cost; and reimplemented SOTA baselines are also sub-cost under our protocol. The contributions are the honest evaluation machinery (date-clustered CIs, realistic costs, cross-period gates, Deflated Sharpe, interaction-information probing) and a precise localization of why *detection is not edge*. We release the harness, the walk-forward sampler, the information-theoretic probes, and the V12.31 expert prompt.

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A V12.31 EXPERT PROMPT

The full ~5,500-character elicited expert prompt (bullish-onset rules C1–C4, bearish asymmetry, disaster composite) is provided in prompts/v12_31_e and reproduced in the supplementary material.